

SYED HUSSAIN ABBAS

Seattle, WA, 98105

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TECHNICAL SKILLS

Languages: React.js, Next.js, Node.js, Python, C/C++, Java, Typescript, NoSQL, SQL, HTML, CSS, Javascript, Express.js, D3.js
Tools: Git, REST APIs, Azure, Jira, Linux/Unix, Selenium/JUnit, AWS/Azure, Docker, Kubernetes, Jenkins, Terraform
AI/ML: OpenCV, Support Vector Machine (SVM), TensorFlow, Scikit-Learn, , GenAI, RAG, Random Forest, NLP
Skills: Full Stack Development, Databases, Unit Testing, Algorithms, DevOps, Agile, CI/CD, UI/UX, Data Visualization

WORK EXPERIENCE

Snohomish County

Seattle, WA

Software Engineer — Seattle University Capstone Project

September 2024 – March 2025

- Design and manage a scalable, centralized data repository using **AWS S3**, **MongoDB**, and **PostgreSQL (PostGIS extension)** to store and organize geospatial and municipal datasets, improving data retrieval efficiency by 60%.
- Enhance data governance and security by implementing **role-based access controls (RBAC)** and **AWS IAM** policies, ensuring compliance with municipal regulations and reducing unauthorized data access incidents.
- Develop an interactive data visualization dashboard using **Tableau** and **Amazon QuickSight**, enabling real-time data insights, cross-departmental collaboration, and improving decision-making efficiency.
- Implement machine learning models in **Python (scikit-learn, TensorFlow)** to detect anomalies in infrastructure data and predict maintenance needs, reducing unexpected system failures by 25% and optimizing budget allocation.

Seattle University

Seattle, WA

Software Engineering Intern

June 2024 – September 2024

- Developed an innovative, production-level software solution for SU's client, *St. Francis House*, optimizing their inventory and asset management system, improving accuracy in asset tracking and operational efficiency by 40%.
- Utilized **Firebase** with **Next.js/React.js** to optimize resource management, achieving a 30% reduction in data retrieval time and a 20% improvement in system scalability by leveraging real-time database, authentication, and cloud storage features.

Stackup Solutions

Lahore, Pakistan

Software Developer

June 2022 – May 2023

- Developed and deployed advanced AI bots, intelligent event-driven automation scripts, and highly dynamic applications to optimize end-to-end workflows, improve user experience (UI/UX), and drive continuous innovation across multiple platforms and business processes.
- Spearheaded the development of an internal AI bot to automate code reviews. Trained the model on 1600+ repos, leveraging **Support Vector Machine (SVM)** and **Random Forest** classification algorithms, resulting in a 45% reduction in manual review time and a 35% decrease in post-deployment issues.
- Implemented a streamlined CI/CD pipeline leveraging **Jenkins** for orchestration, **AWS** services for scalable infrastructure provisioning, and **Terraform** for infrastructure as code management, resulting in a 30% reduction in deployment time and a 41% decrease in manual errors.
- Redesigned the distributed architecture of a computer vision platform, transitioning from **microservices** to a **macroservices** model, reducing latency by 38% and operational costs by 65% while improving system reliability and fault-tolerance.

EDUCATION

Seattle University

Expected Graduation: March 2025

Master of Science in Computer Science - SWE Specialization

- GPA:** 3.81/4.0.
- Coursework:** Software Engineering, Artificial Intelligence, Cloud Computing, Software Architecture, Software Project Management, Software Testing, Machine Learning, Computer Networks.

PROJECTS

Bliss Bazaar | Next.js, Node.js, Typescript, Prisma ORM, MongoDB

February 2024

- Built an e-commerce platform with personalized recommendations for enhanced online shopping experiences.
- Integrated **OAuth** authentication with **SHA-256** hashing to fortify transaction security and protect sensitive data, reducing vulnerability by 32% through encrypted credential storage and secure access management.

Rafah | React.js, Node.js, Express.js, Flask, Python, MongoDB, Scikit-Learn

June 2023

- Engineered a digital gateway to connect students with NGOs for documented volunteer work, facilitating graded reflections used by educational institutes to waive community service-based credit hours accurately.
- Integrated a content-based recommendation system for student-NGO matching, leveraging **Support Vector Machines (SVM)** to learn preferences from student profiles and **Cosine Similarity index** to find best-fit NGOs based on mission, interests, and goals.
- Evaluated model predictions using **Confusion Matrix** and **ROC-AUC** to fine-tune the model hyperparameters, achieving 78% operational accuracy.